Register Circumfixes overwrite Downstep in Hausa and Chimiini

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Main Claim

Downstep Deletion in interrogative utterances can be modelled as circumfixation of h-register features.

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Introduction

- Global phonological processes, such as phrasal accent, have been a problem for generative phonology with a strict concatenative view.
- Q-Raising/Downstep Deletion in Hausa and Chimiini is one such process.
- I argue these can be derived by circumfixing high register features to the domain of downstep.

The Erasure Problem (Trommer, 2011)

- Affixal registers behave different from underlying one.
- Affixed registers spread through a whole domain whereas underlying register feature do not.
- Solution: Contiguity constraint (cf. Trommer 2011) makes reference to both underlying and specifically to surface structure.

(1) Contiguity(register) (=Cont(r)):
Count one violation for every tonal root node $\circ_1$, such that

a. $\circ_1$ is visibly associated to a register r_1 of color c_1 and
b. r_1 is both preceded and succeeded by some register of color c_2.

Data

Automatic and Non-Automatic Downstep

Automatic downstep

- Automatic downstep is the lowering of the register after low tones.
- All following high tones will be slightly lowered.

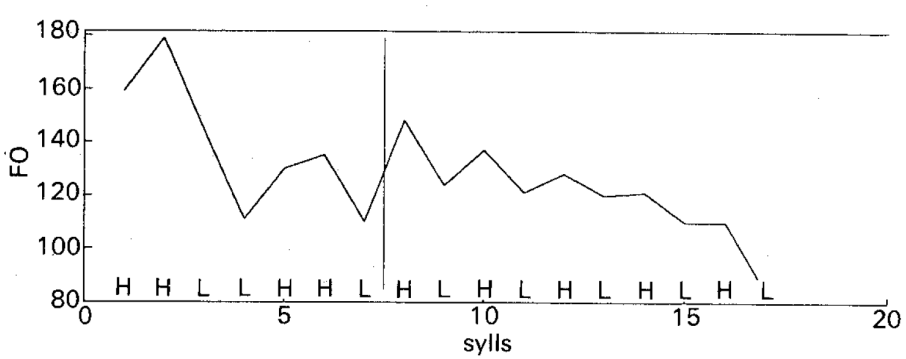
Non-Automatic downstep

- Non-Automatic downstep is the lowering of the register after high tones.
- All following high tones will be slightly lowered.

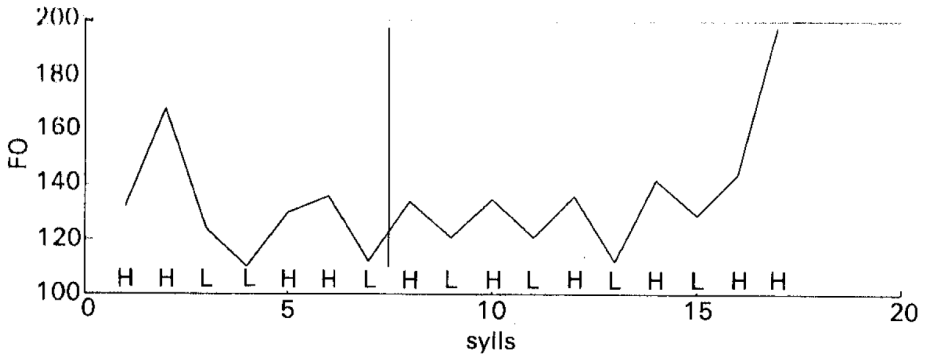
Downstep Deletion in Hausa

- In Hausa, automatic downstep is deleted in the last phonological phrase of a polar question.
- Otherwise, high tones following low tones are downstepped in all contexts.

Downstep Deletion in Hausa II



Yaa aikà wà Manii)$_{\varphi}$ làabaarìn wannàn yaaròn alàrammà



Yaa aikà wà Manii)$_{\varphi}$ làabaarìn wannàn yaaròn alàrammà

Figure 1: No Downstep in polar questions in Hausa (Inkelas & Leben, 1990, 22-23)

Downstep Deletion in Chimiini

- In Chimiini, non-automatic downstep is deleted in the whole utterance of polar questions.
- Otherwise, consecutive high tones are downstepped.

Downstep Deletion in Chimiini II

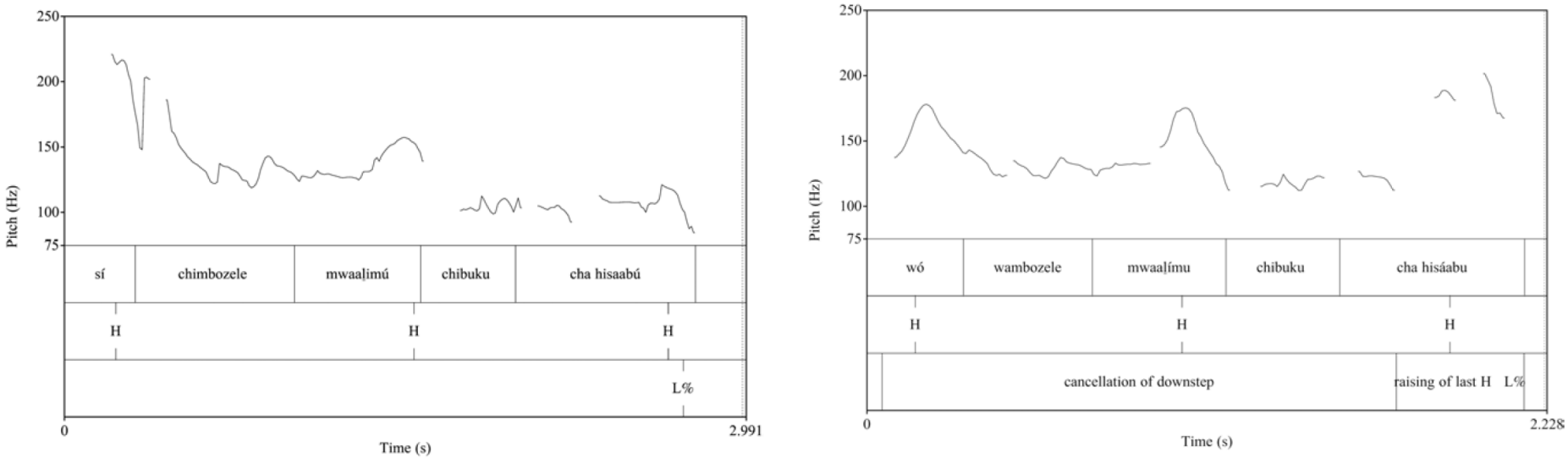


Figure 2: No Downstep in polar questions in Chimiini (Kisseberth, 2017, 235-237)

Analysis

Analysis

Basic Idea:

- A high register circumfix attaches to the domain of downstep deletion in questions.
- A CONTIGUITY constraint forces these registers to overwrite intervening high registers.
- Since the tonal root nodes are occupied, docking/spreading of low register is blocked.
- Hence, downstep is blocked in questions.

Theoretical Background I

Colored Containment (van Oostendorp, 2007; Revithiadou, 2007)

- Containment OT system: input has to be contained in each candidate.
- Instead of deletion, association lines are marked as phonetically invisible.
- The only morphological information that is available to the phonology are the differences in morphological affiliation (=color).
- Epenthetic material has no morphological affiliation (=colorless).

\+ Constraints can make reference to the surface structure and/or complete structure.

Theoretical Background II

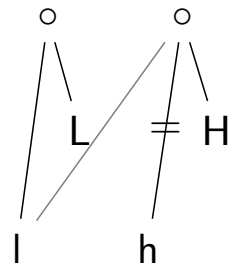
Register Tier Theory (Snider, 1990)

- Autosegmental division between tones and registers.
- Register features h and l on one tier.
- Tone features H and L on another tier.
- Both linked to tonal root nodes (○).
- Register features lower/raise pitch globally.
- Tonal features lower/raise pitch locally.

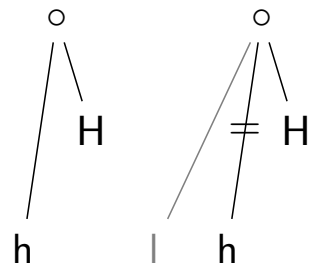
Theoretical Background III

(2) Representation of Downstep in the Register Tier Theory

a. Automatic Downstep



b. Non-Automatic Downstep



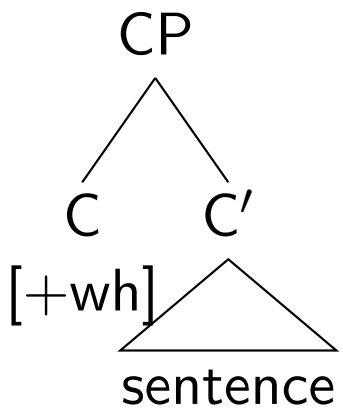
- I assume SHARE(low) and OCP(high) as the trigger of the two types of downstep.

Theoretical Background IV

Pivot Theory of Affixation (Fitzpatrick, 2004; Yu, 2007)

- Affixes are specified for the prosodic constituent and the position they attach to.
- \+ Extension: Affixes can also take large hosts like intonational phrases and phonological phrases.
- One part of the circumfix is prefixed to the left edge of the phrase and the other part is suffixed to the right edge of the same phrase (cf. Trommer 2011).

(3) Syntactic structure of a question



Theoretical Background V

(4) Constraints triggering downstep

a. SHARE(low) (=SH(l))
Count one violation for every pair of tonal root nodes, such that one of them is associated to a low register and the other one is not associated to that very same low register.

b. OBLIGATORYCONTOURPRINCIPLE(high) (=OCP(h))
Count one violation for every pair of adjacent high registers on the register tier.

(5) Constraints on Tonal Autosegmental structures

a. ASSOCIATE(N) (=AS(N))
Count one violation for every node N that is not associated to a tonal root node.

b. *F,G
Count one violation for every tonal root node that is associated to a node F and a node G.

Theoretical Background VI

\+ NoLineCrossing is violable (cf. Bagemihl 1989).

(6) NoLineCrossing(register)(=NLC(r))
Count one violation for every ordered pair of register features r_1 and r_2 which are linked to tonal root nodes $\circ_1$ and $\circ_2$ such that

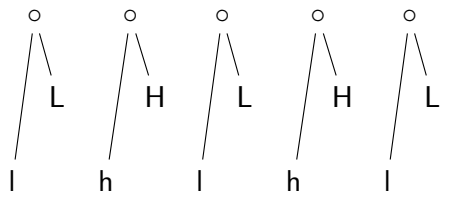
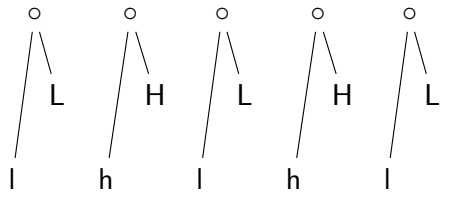
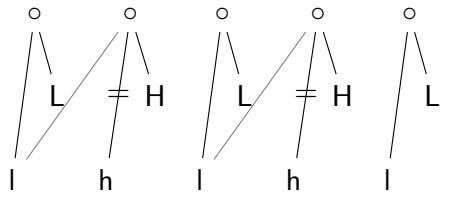
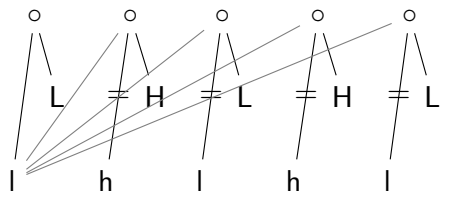
a. r_1 is linked to $\circ_1$ and
b. r_2 is linked to $\circ_2$ and
c. r_1 precedes r_2
d. but $\circ_1$ succeeds $\circ_2$.

Automatic Downstep in Hausa I

- Automatic Downstep in Hausa is triggered by a SHARE(low) constraint.
- Further spreading of underlying registers incurs fatal violation of ASSOCIATE(low), since more low registers would have to be delinked.

Automatic Downstep in Hausa II

(7) Evaluation of [. . . ròn ˈálàˈrámmà] – downstep

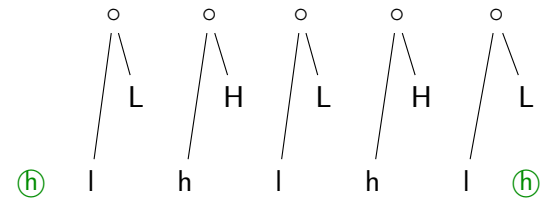
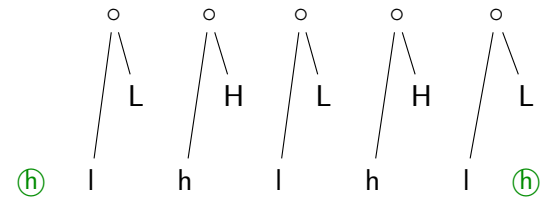
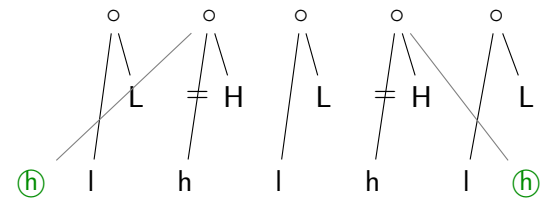
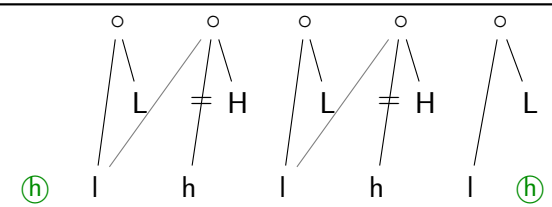
Input: 	As(l)	NLC(r)	SH(l)	As(h)	*H, l
a. 			***!*		
☞ b. 			**	**	**
c. 	*!*			**	**

Downstep Deletion in Hausa I

- In questions, a high register cirumfix attaches to the last prosodic phrase of the utterance (ⓗ- -ⓗ).
- A high ranked Contiguity constraint forces this circumfix to overwrite as many intervening high registers as possible.
- It is not possible to overwrite low registers since Associate(low) is undominated.
- Automatic downstep is blocked, since low register spreading would keep the high register cirumfix from docking – a violation of Contiguity.

Downstep Deletion in Hausa II

(8) Derivation of [. . . ròn álàrámmà?] – no downstep

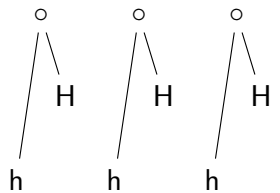
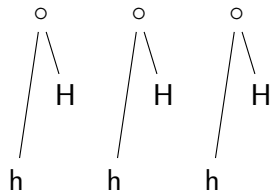
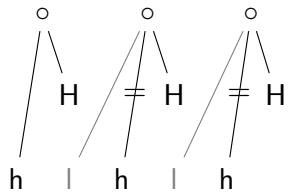
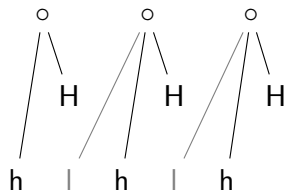
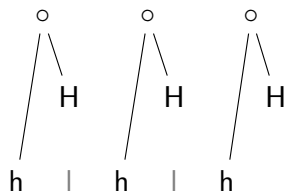
		As(l)	Cont(r)	NLC(r)	Sh(l)	As(h)	*H,l
Input:							
a.			****!*		****	**	
☞b.			***	**	****	**	
c.			****!*		**	****	**

Non-Automatic Downstep in Chimiini

- Downstep in declarative sentences in Chimiini consists of epenthetic low registers.
- These are inserted due to an OCP(high) constraint.
- They have to be associated, due to a high ranked ASSOCIATE(low) constraint.
- High registers are then delinked due to a high ranked constraint against multiple registers associated to one tonal root node.

Non-Automatic Downstep in Chimiini

(9) Evaluation of a declarative sentence in Chimiini – downstep

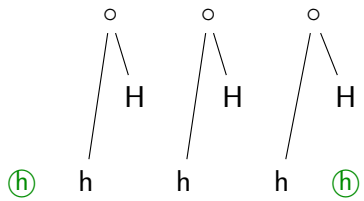
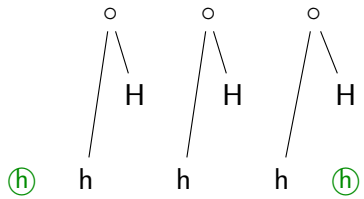
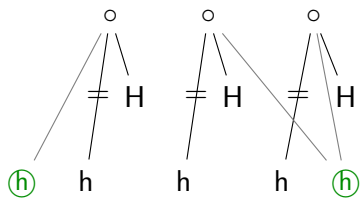
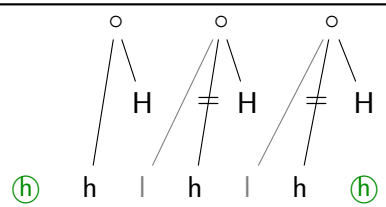
	*r,r	As(l)	OCP(h)	DEP(l)	As(h)
Input: 					
a. 			*!*		
☞ b. 				**	**
c. 	*!*			**	
d. 		*!*		**	

Non-Automatic Downstep in Chimiini

- In questions, doenstep is deleted – again – due to a high register circumfix.
- These are forced to overwrite intervening registers due to a Contiguity(register) constraint.
- Downstep is blocked again, since inserting and associating low registers would intervene between the two registers.
- This would fatally violate the Contiguity constraint.

Downstep Deletion in Chimiini

(10) Evaluation of a question in Chimiini – no downstep

		*r,r	Cont(r)	As(l)	OCP(h)	DEP(l)	As(h)
Input:							
a.			*!**		****		**
☞b.					****		***
c.			*!**		**	**	****

Discussion

Discussion

Alternative approaches to overwriting

- Cophonology Theory (Orgun, 1996) easily derives global downstep deletion.
- Deletion of downstep only in the last phonological phrase would require a constraint explicitly refering to this domain.

Alternative representations of downstep

- Downstep is often represented as a floating low tone (i.a. Paster & Kim 2011)
- This would either require a process of floating tone insertion in statements or a floating tone deletion rule in questions.
- Both are difficult to model.

Conclusion

Conclusion

- It is possible to derive global phrasal accent phenomena – like downstep deletion – in a purely concatenative way.
- Register circumfixes are forced to overwrite intervening registers due to a high ranked Contiguity constraint.

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